AcrossPlace: Do Vision–Language Models Recognize Places Beyond Their Source Views? A Comprehensive Benchmark

Anonymous CVPR submission

Abstract

*Can language identify a place when the image used to describe it is unavailable? This question demands distinguishing place identity from source-image agreement, while controlling for geographic partitions, candidate-gallery size, and annotation quality. We present **AcrossPlace**, a provenance-aware benchmark for bidirectional language–vision place recognition. Built on GSV-Cities, it comprises **56,726 places across 23 cities**, **156,800 automatically generated, rule-filtered descriptions**, and **113,404 target images**. Source images are excluded from principal target galleries, and text-to-image (T2I) and image-to-text (I2T) retrieval share a place-level relevance criterion. Two fixed protocols evaluate unseen places in represented cities (P0) and places in held-out cities (P1). Paired source/target comparisons, gallery-matched controls, fixed-query scaling, and a blinded text audit isolate complementary evaluation factors. Across three supervised training seeds, supervised SigLIP2 T2I Recall@1 decreases from **19.69%** to **12.31%** when source images are replaced by independent observations in a controlled P0 test. On P1, enlarging the gallery from 500 to 8,425 places reduces fixed-query recall from **14.67%** to **2.87%**. A 240-description audit further shows that rule-based acceptance does not establish factual correctness. These findings expose limits of aggregate retrieval scores and establish AcrossPlace as a reproducible framework for assessing observation transfer, geographic evaluation, and language evidence in place recognition.*

1. Introduction

A place has a geographic identity that persists across images, yet its visible evidence does not. A description grounded in one observation may match that image while failing to identify another observation of the same location. This distinction is central to language–vision place recognition: *retrieving the image that supports a caption is not equivalent to retrieving the place it describes*. The challenge becomes more pronounced when candidate galleries contain visually similar locations or when evaluation moves to different cities.

Recent studies have advanced language-based place retrieval through multi-view matching [6], bidirectional multimodal representations [11], and large-scale linguistic supervision [3]. These advances sharpen a fundamental question: **To what extent do language–vision models recover place identity from visual evidence independent of the description source?** Answering it requires more than a place-level accuracy score. The score also depends on which observation is retrieved, which competing locations populate the gallery, and whether the description is both grounded and sufficiently distinctive. Conflating these factors can obscure the capability a benchmark is intended to measure.

We present **AcrossPlace**, an evaluation standard that connects *traceable language evidence*, *bidirectional place retrieval*, and *controlled changes in retrieval conditions*. Built on GSV-Cities [1], AcrossPlace comprises **56,726 places across 23 cities**, **156,800 rule-filtered descriptions**, and **113,404 target images**. Each description is linked to the observation from which it was generated; principal retrieval instead uses disjoint target observations. Both text-to-image (T2I) and image-to-text (I2T) retrieval are scored by shared place identity, allowing multiple relevant gallery samples without requiring an exact image–caption pair.

Two fixed protocols organize geographic evaluation. **P0** tests unseen places in represented cities, whereas **P1** tests places in held-out cities. Their principal scores are complemented by paired source–target comparisons, cardinality-matched geographic galleries, and nested gallery scaling with fixed queries. A blinded audit separately measures whether captions are supported by their source observations, visible in the target set, and specific to a place. Together, these conditions distinguish changes in the evidence being matched from changes in the retrieval problem itself.

The resulting measurements are consequential. For supervised SigLIP2 on P0, replacing source images with independent target observations reduces paired T2I Recall@1 from **19.69% to 12.31%**. For a fixed P1 query cohort, expanding the gallery from 500 to 8,425 places reduces Recall@1 from **14.67% to 2.87%**, without changing the queries or model. Among 240 audited descriptions, 166 are

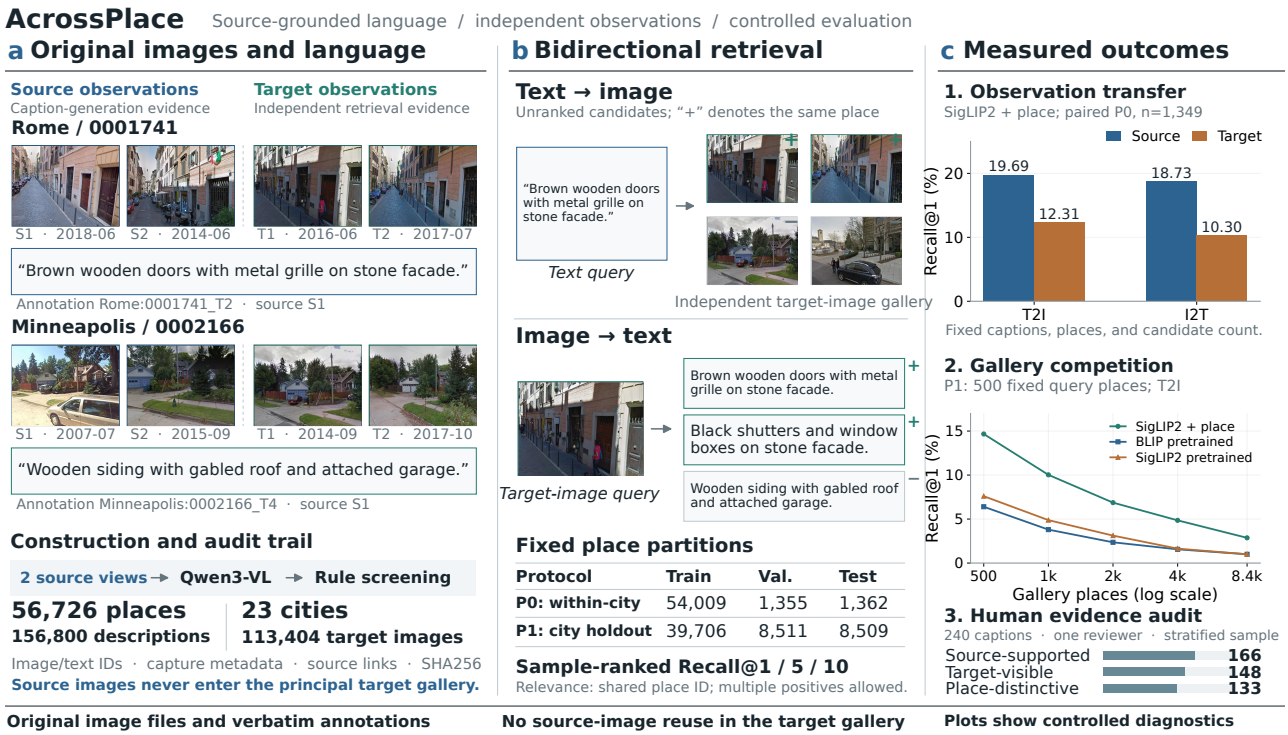


Figure 1. AcrossPlace overview using original images and verbatim annotations. (a) Two places with source/target dates and traceable text. (b) Bidirectional place relevance, illustrated with unranked candidates. (c) Archived paired-P0 and fixed-query-P1 diagnostics, distinct from principal full-gallery scores; supervised results average three seeds. Human counts describe a stratified, single-reviewer audit.

fully source-supported and 148 have sufficient target evidence; rule-based acceptance alone does not reliably rank factual correctness. Thus, observation transfer, candidate competition, and language evidence materially shape the interpretation of place-recognition scores.

Our contributions are threefold:

1. **A provenance-grounded benchmark resource.** AcrossPlace links language to its generating observations and separates these from retrieval targets, establishing a place-identity-based evaluation unit across both retrieval directions.
2. **A reproducible evaluation standard.** Fixed P0/P1 partitions and bidirectional protocols are coupled with paired observation transfer, gallery-matched comparison, and fixed-query scaling to expose distinct sources of evaluation variation.
3. **A diagnostic empirical account.** Common-budget pretrained and adapted references, together with a three-axis human audit, quantify how retrieval conclusions change with visual evidence, gallery composition, and annotation quality.

2. Related Work

2.1. Language and multimodal place recognition

GSV-Cities provides geographically diverse images and place supervision [1]. Text4VPR/Street360Loc pairs multi-directional views with natural-language descriptions [6]; UniLoc evaluates image, text, and point-cloud retrieval in both directions [11]. LaVPR introduces large-scale language annotations and already evaluates descriptions against reference images distinct from their sources [3]. DialogueVPR studies interactive rather than static language queries [8]. AcrossPlace turns these established retrieval settings into a common diagnostic standard, connecting description provenance and bidirectional place relevance with paired observation, geographic, gallery-scale, and language-evidence analyses.

2.2. Diagnostic benchmark design

An informative benchmark must specify which capability a retrieval score measures. Question-driven evaluations such as RealUnify [7] illustrate the value of diagnostics alongside aggregate scores. AcrossPlace applies this principle to place identity by jointly specifying observation provenance, geographic partitions, gallery composition, and language evidence. Table 1 summarizes the documented evaluation

protocols of related resources.

2.3. Retrieval reference models

CLIP [5] and SigLIP2 [9] learn shared visual–textual embeddings; BLIP supports contrastive features and image–text matching [4]. We combine established place-supervised losses with LoRA [2] and Multi-Similarity learning [10] to build reference systems. Our comparison uses a common adaptation budget to probe benchmark sensitivity, not to claim native-schedule reproduction or a definitive model ranking.

3. AcrossPlace Benchmark

AcrossPlace operationalizes a single principle: *the identity of a place must be evaluated independently of the observation used to describe it*. The benchmark therefore specifies three linked elements: a traceable description source, a disjoint retrieval target, and a place-level relevance rule. Geographic partitions and controlled diagnostics then expose how the resulting scores vary with observation transfer, candidate competition, and text evidence (Fig. 1).

3.1. Resource construction and observation provenance

Source and target observations. We start with 62,279 GSV-Cities places, each represented by four images (249,116 images in total). Qwen3-VL-4B-Instruct generates natural-language descriptions from designated *source observations*; different images of the same place constitute the *target pool*. Each description preserves its source-image link. Image and text identifiers, canonical paths, capture time, GPS metadata, panorama identifiers, and SHA256 hashes are retained in the manifests, making source–target separation auditable rather than implicit.

Eligible resource. Rule-based screening produces **56,726 places across 23 cities**, with **156,800 accepted descriptions** and **113,404 target images**. Accepted descriptions form the principal query and text-gallery pool; rule-uncertain descriptions are retained for separate quality diagnostics. The screening labels indicate compliance with the generation rules, not verified semantic truth. All reported benchmark totals refer to the eligible resource rather than the initial selection.

Independent visual evidence. Source and target observations differ in capture metadata and image identity. Median source–target heading differences are 10° in P0 and 9° in P1; median capture-time gaps are 46 and 51 months, respectively. No source–target pair in the controlled observation diagnostic is byte-identical or shares a panorama ID. Accordingly, *cross-view* denotes distinct visual observations of the same place; it does not require a large angular change or guarantee that every described landmark remains visible.

3.2. Geographic evaluation protocols

The benchmark defines two fixed, complementary partitions of the same eligible resource (Table 2). Both use place-disjoint splits and preserve predetermined query and gallery manifests.

P0: unseen places within represented cities. Training, validation, and test splits contain 54,009, 1,355, and 1,362 places, respectively. Their description and target-image counts are reported in Table 2. The original formal split is held fixed for evaluation.

P1: transfer to held-out cities. Cities are disjoint across splits: four are assigned to validation and four distinct cities to testing. Training, validation, and test splits contain 39,706, 8,511, and 8,509 places. City assignment is determined by metadata and a fixed seed, independently of model performance.

P0 measures generalization to new places within a represented city distribution; P1 measures performance under a city-disjoint evaluation setting. Because the partitions also differ in training size, query identities, and candidate-gallery composition, their raw score gap measures the *combined protocol difference*. Cardinality-matched controls below provide a more focused comparison of geographic settings.

3.3. Bidirectional retrieval and place-level relevance

Let $p(x)$ denote the place identity of an image or description x. In **text-to-image (T2I)** retrieval, a description queries the target-image gallery; in **image-to-text (I2T)** retrieval, a target image queries the description gallery. Both directions share the same relevance rule: a candidate g is relevant to query q if $p(g) = p(q)$, irrespective of their direct image–caption association. This permits multiple positive candidates per place.

We rank *individual samples*, not deduplicated places, and report Recall@K for $K \in \{1, 5, 10\}$:

$$R@K = \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}[\exists g \in \mathrm{TopK}(q) : p_g = p_q] \,. \quad (1)$$

Repeated items from one place retain separate gallery ranks. Principal galleries comprise the corresponding held-out split and never include caption-source images as retrieval targets. GPS coordinates and source images are used to construct and audit the benchmark, not as model inputs at inference; I2T retrieves stored descriptions rather than generating new ones.

Place-ID relevance supplies a stable retrieval target, while language itself may remain ambiguous: different places can share a plausible description, and a source-supported fact can be absent from a target view. We measure these semantic limitations independently through the quality audit in Sec. 5.3, without changing the official relevance labels after observing predictions.

Table 1. Published language–vision place-retrieval protocols. Counts refer to different units; NE denotes information not established by the cited source, not absence of a capability. AcrossPlace integrates provenance, bidirectional retrieval, paired observations, gallery controls, and a three-axis text audit.

Resource	Scale (unit)	Query / directions	Observation relationship	Geographic protocol	Quality evidence
Street360Loc [6]	7k locations	View groups; T2I	Four panorama directions	Location split	Generated, proofread
UniLoc [11]	29,438 triplets	Image/text; both	Image-centric triplets	Sequence split	Semantic-label text
LaVPR [3]	651,865 pairs	Language; T2I	Source query excluded	Existing VPR splits	Grounding and inspection
DlgQuest-Cities [8]	106,880 images	Multi-turn dialogue	Region and dialogue evidence	City-holdout NE	Screening and inspection
AcrossPlace	56,726 places	Single sample; both	Explicit provenance; paired controls	P0/P1; scale controls	240-item, three-axis audit

Table 2. Fixed geographic partitions. P0 and P1 use the same 56,726 places, 156,800 valid descriptions, and 113,404 target images, but assign places to different splits. Source observations are kept separate.

Protocol	Split	Places	Texts	Targets
P0	Train	54,009	149,126	107,970
P0	Validation	1,355	3,876	2,710
P0	Test	1,362	3,798	2,724
P1	Train	39,706	109,809	79,390
P1	Validation	8,511	23,622	17,000
P1	Test	8,509	23,369	17,014

3.4. Controlled diagnostic evaluations

Principal Recall@K reports performance under fixed protocols. Three complementary diagnostics determine *which evaluation factor drives a change in that score*.

D1: Observation transfer. We form one-source/one-target pairs using one accepted, single-source description per eligible place. The description and place cohort are fixed while the candidate gallery switches between source images and independent target images at equal cardinality. For I2T, the paired source or target image serves as the query against the corresponding caption set. The resulting singleton cohorts contain 1,349 P0 and 8,425 P1 places; 13 and 84 test places, respectively, do not satisfy the single-source eligibility rule. This isolates an observation replacement within the controlled cohort, rather than any one of its possible causes (time, orientation, or occlusion).

D2: Geographic composition and gallery size. To compare P0 and P1 with matched candidate counts, we sample five 1,000-place galleries per protocol, retaining one image and one description per place. Separately, 500 query places and their queries are fixed while nested candidate pools grow through 500, 1,000, 2,000, 4,000, and the complete eligible pool where available. All pools are specified in advance and contain their query-place positives; candidate lists are never corrected after retrieval. Gallery draws and training seeds are treated as distinct sources of variation.

D3: Language evidence and screening. In P1, 7,735 test places have both rule-accepted and rule-uncertain descriptions. A matched subset of 4,291 places additionally controls for description type, source observation, and word-count bin. The query-place cohort, candidate count, and background captions are fixed while cohort descriptions vary; the P0 test split has no corresponding uncertain-caption pool. A separate blinded audit covers 240 descriptions (120 per rule-status stratum), judging source correctness, target visibility, and place specificity independently. These measurements expose screening and evidence differences without treating a rule label as a human correctness judgment.

4. Evaluation Setup

4.1. Reference systems and adaptation

We test four pretrained controls—CLIP ViT-B/16, CLIP ViT-B/32, BLIP contrastive features, and SigLIP2—and four supervised recipes: CLIP with place loss; a LaVPR-inspired BLIP adaptation with Multi-Similarity (MS) loss; SigLIP2 with MS loss; and SigLIP2 with bidirectional place loss. Four SigLIP2 variants further isolate residual heads and vision-only versus text-only adaptation. Each supervised configuration is trained with seeds 42, 43, and 44.

All recipes use a *three-epoch common budget*. The standard setup samples 16 places per batch, at most two target images and four descriptions per place; uses AdamW; and applies rank-eight LoRA to query/value projections in the final four blocks of both encoders. Residual heads start as identity mappings. The BLIP+MS reference instead uses all-linear rank-64 rank-stabilized LoRA, an image-to-text objective, and scaled SGD. This equal-duration compari-

son does not imply convergence or reproduce the original methods' full training schedules; SALAD is not included.

4.2. Checkpoint selection and statistical reporting

Checkpoints are selected using the *weaker* validation-direction Recall@100, breaking ties by weaker-direction Recall@1. A frozen BLIP image–text matching (ITM) model reranks coarse Top-100 results; its score is fused with coarse similarity using a weight chosen on a fixed validation grid. Model identities, checkpoint, fusion weight, and evaluation configuration are frozen before test inference. Coarse and reranked systems are reported separately.

Supervised results report training-seed variability; pretrained models use one frozen run. Gallery repetitions quantify candidate-sampling variability, and source/target confidence intervals resample paired places. Audit subgroups are descriptive because they contain only 240 captions, repeated places, and a single annotator.

5. Results and Diagnostic Analysis

We investigate three factors obscured by aggregate Recall@K: transfer beyond caption-source images, competition within candidate galleries, and the reliability of language evidence. Principal P0/P1 scores first establish reference performance; controlled tests then isolate what each score measures. Training-seed SD, gallery-draw variability, and place-bootstrap intervals represent distinct uncertainties.

Reference performance. Tables 3 and 4 show that supervision substantially improves the tested baselines. On P0, SigLIP2 T2I/I2T R@1 increases from 4.16%/5.29% pretrained to 12.95%/13.64% place-supervised, reaching 15.24%/17.35% with validation-selected ITM reranking. BLIP+MS achieves the highest observed P0 coarse scores (17.54%/23.62%) among tested configurations, but reranking decreases them to 16.87%/21.28%. On P1, BLIP+MS reaches 6.09%/8.36% coarse and 6.23%/8.18% reranked R@1. Thus, reranking benefits depend on the representation; coarse and reranked results remain separate fixed systems.

Training scope. All supervised references receive three epochs. Validation performance continues to increase over this budget; for BLIP+MS on P0, T2I/I2T R@1 rises from 13.97%/20.26% to 16.94%/23.75%. These runs define a common-budget comparison, not converged performance, and losses from different objectives are not directly comparable.

5.1. Finding I: Source-image agreement overstates observation transfer

To distinguish caption-source matching from place recognition, we retrieve identical captions against equal-sized galleries containing either their source images or independent target observations (Fig. 2, Table 5). Place cohorts are fixed, with one image and one description per place.

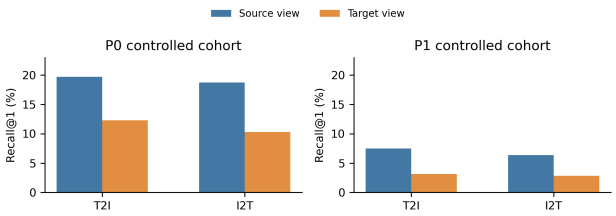


Figure 2. Observation-transfer diagnostic. Identical captions and place cohorts are retrieved against equal-size source or independent-target galleries. Bars average three supervised SigLIP2 seeds; this one-sample-per-place setting differs from principal retrieval.

For place-supervised SigLIP2 on P0, T2I R@1 decreases from 19.69% to 12.31%, and I2T from 18.73% to 10.30%. Paired place-bootstrap 95% intervals for target-minus-source differences are $[-9.24, -5.49]$ and $[-10.16, -6.67]$ percentage points. On P1, T2I declines from 7.49% to 3.11% and I2T from 6.35% to 2.85%, with intervals $[-4.92, -3.84]$ and $[-4.01, -3.01]$. Pretrained BLIP also declines, although the P0 pretrained SigLIP2 T2I interval $[-2.15, +0.52]$ includes zero.

These paired results establish a measurable *observation-transfer gap* for several references. They do not isolate viewpoint, capture time, occlusion, or visibility individually; confidence intervals reflect place resampling conditional on frozen models, not seed variation. Source-image retrieval is therefore an incomplete proxy for identifying a place from independent evidence.

5.2. Finding II: Gallery competition alters geographic comparisons

P0 and P1 jointly vary city coverage, training cohorts, queries, and gallery size. We first equalize candidate counts, then expand nested galleries with fixed queries to separate retrieval scale from these broader protocol changes.

At 1,000 places (Table 6), place-supervised SigLIP2 attains 13.58%/11.69% T2I/I2T R@1 on P0 versus 10.61%/9.37% on P1, narrowing the full-gallery gap. The pattern is not universal: pretrained BLIP T2I is 3.88% on P0 versus 4.12% on P1. Equal cardinality does not equalize query identities, training data, or distractor semantics.

In the fixed-query P1 experiment (Fig. 3), increasing the gallery from 500 to 8,425 places lowers supervised SigLIP2 T2I R@1 from 14.67% to 2.87% and I2T from 14.73% to 3.13%. Intermediate T2I values are 10.03%, 6.87%, and 4.85% at 1,000, 2,000, and 4,000 places. Queries, positive place identities, and weights remain fixed. This expected effect of additional candidates demonstrates why full-gallery P0–P1 differences cannot be attributed solely to geographic

Table 3. P0 test Recall@K (%). Supervised R@1: mean ± SD across three training seeds; R@5/10: seed means. Pretrained controls use one frozen checkpoint. ITM denotes frozen BLIP matching of coarse Top-100 results. MS models are common-budget adaptations, not reproductions of native training schedules.

Reference	Scoring	T2I			I2T		
		R@1	R@5	R@10	R@1	R@5	R@10
CLIP B/32 (zero-shot)	Coarse	1.76	5.87	9.35	2.68	8.48	12.74
CLIP B/16 (zero-shot)	Coarse	2.13	5.61	8.77	3.63	9.58	13.62
BLIP (zero-shot)	Coarse	4.08	10.01	13.72	4.41	11.01	16.04
SigLIP2 (zero-shot)	Coarse	4.16	9.06	12.59	5.29	11.34	15.93
SigLIP2 (zero-shot)	+ ITM	7.61	16.06	20.30	10.02	19.42	24.01
CLIP B/16 + place loss	Coarse	10.59 ± 0.12	23.98	33.39	11.17 ± 0.11	26.35	36.81
CLIP B/16 + place loss	+ ITM	13.54 ± 0.29	27.08	36.40	15.38 ± 0.04	31.64	40.79
BLIP + MS adaptation	Coarse	17.54 ± 0.30	35.13	44.89	23.62 ± 0.27	42.73	52.80
BLIP + MS adaptation	+ ITM	16.87 ± 0.28	31.60	41.11	21.28 ± 0.20	40.44	49.47
SigLIP2 + MS	Coarse	14.25 ± 0.11	29.31	38.05	17.35 ± 0.09	36.23	45.69
SigLIP2 + MS	+ ITM	14.56 ± 0.03	29.47	37.34	19.20 ± 0.10	36.28	45.33
SigLIP2 + place loss	Coarse	12.95 ± 0.22	27.68	36.69	13.64 ± 0.27	30.87	41.45
SigLIP2 + place loss	+ ITM	15.24 ± 0.30	29.73	38.12	17.35 ± 0.15	34.88	44.77

Table 4. P1 full-gallery test Recall@K (%). Supervised R@1: mean ± SD over three seeds; R@5/10: means. Reranking weights are validation-selected. P1 differs from P0 in training and query cohorts as well as gallery composition and size.

Reference	Scoring	T2I			I2T		
		R@1	R@5	R@10	R@1	R@5	R@10
CLIP B/32	Coarse	0.57	1.70	2.66	0.81	2.42	3.79
CLIP B/16	Coarse	0.42	1.51	2.46	0.92	2.66	4.08
BLIP	Coarse	1.14	3.30	4.97	1.10	3.27	4.91
SigLIP2	Coarse	1.20	3.54	5.23	1.42	3.72	5.30
SigLIP2	+ ITM	2.75	6.35	8.60	2.66	6.49	8.80
CLIP + place	Coarse	2.70 ± 0.08	7.80	11.60	2.82 ± 0.05	8.02	11.90
CLIP + place	+ ITM	4.31 ± 0.05	10.34	14.23	4.70 ± 0.03	11.54	15.82
BLIP + MS	Coarse	6.09 ± 0.12	14.30	19.66	8.36 ± 0.10	18.23	24.38
BLIP + MS	+ ITM	6.23 ± 0.02	13.87	19.00	8.18 ± 0.09	17.63	23.68
SigLIP2 + MS	Coarse	4.11 ± 0.04	10.31	14.84	5.18 ± 0.04	12.91	18.10
SigLIP2 + MS	+ ITM	5.25 ± 0.04	11.92	16.47	6.40 ± 0.04	14.79	20.07
SigLIP2 + place	Coarse	3.68 ± 0.05	9.52	13.74	3.46 ± 0.04	9.97	14.59
SigLIP2 + place	+ ITM	5.21 ± 0.05	12.03	16.47	5.72 ± 0.08	13.38	18.25

Table 5. Source-to-target observation transfer (%; Δ in percentage points). Same captions and equal-size galleries for 1,349 P0 and 8,425 P1 places. 95% CIs use 2,000 paired place bootstraps after averaging seed outcomes; intervals exclude training-seed uncertainty.

Protocol	Reference	T2I R@1			I2T R@1		
		Source	Target	Δ [95% CI]	Source	Target	Δ [95% CI]
P0	BLIP (pretrained)	7.78	3.48	-4.30 [-5.78, -2.82]	6.89	3.63	-3.26 [-4.52, -2.00]
P0	SigLIP2 (pretrained)	5.71	4.89	-0.82 [-2.15, +0.52]	6.08	4.60	-1.48 [-2.82, -0.22]
P0	SigLIP2 (place-supervised)	19.69	12.31	-7.39 [-9.24, -5.49]	18.73	10.30	-8.43 [-10.16, -6.67]
P1	BLIP (pretrained)	2.85	1.12	-1.73 [-2.12, -1.35]	2.49	1.04	-1.45 [-1.82, -1.09]
P1	SigLIP2 (pretrained)	2.99	1.27	-1.72 [-2.10, -1.36]	2.56	1.13	-1.44 [-1.76, -1.10]
P1	SigLIP2 (place-supervised)	7.49	3.11	-4.37 [-4.92, -3.84]	6.35	2.85	-3.50 [-4.01, -3.01]

shift.

5.3. Finding III: Rule acceptance does not certify place evidence

Place-ID correctness does not guarantee that a retrieved description is factually supported or distinctive. We audit 240 descriptions from 237 P1 places, equally stratified by rule-valid and rule-uncertain status. A single blinded reviewer independently judges source correctness, target visibility, and place specificity (Table 7). Of 240 captions, 166 are fully source-supported, 148 have sufficient target evidence, and 133 are distinctive; partial and unknown labels remain separate from contradiction. Full source support is observed for 81/120 rule-valid and 85/120 rule-uncertain descriptions, providing no reliable factual-quality ordering from automated status alone.

In a matched 4,291-place P1 comparison, rule-valid captions yield higher supervised SigLIP2 R@1 than uncertain captions: 3.16% versus 2.19% for T2I and 3.25% versus 1.78% for controlled I2T. These differences concern retrieval, not verified truth. Human-audit subgroups show supervised T2I R@10 of 17.79% for target-visible versus 0.00% for not-visible captions, and 17.07% for source-supported versus 4.35% for contradicted captions. Small, overlapping groups and one annotator preclude causal or population-level inferences.

The audit uses 17,014 target images for T2I; controlled I2T uses one caption per place with focal replacement in an 8,509-caption gallery. These conditions differ from principal all-description I2T retrieval. Retrieval errors illustrate semantic aliasing: plausible matches can share facades, vegetation, or entrances yet refer to different places. Together, the evidence separates automatic acceptance, visual support, specificity, and place-ID retrieval success.

6. Discussion and Conclusion

What AcrossPlace measures. The three findings identify distinct evaluation effects. Agreement with a caption-source image can exceed recognition from an independent observation; candidate-gallery size substantially affects recall; and automated text acceptance cannot substitute for verifying linguistic evidence. AcrossPlace makes these factors explicit through provenance-aware bidirectional retrieval, paired observation controls, fixed-query gallery scaling, and independent caption auditing.

Implications. Benchmark comparisons should report both retrieval directions, actual gallery sizes, and the fixed query/gallery manifests. Paired source–target tests assess observation transfer, while gallery-controlled experiments constrain the interpretation of geographic comparisons. P0–P1 differences remain *protocol effects*, not estimates of city shift alone, because the training and test cohorts also differ. Similarly, model-dependent reranking gains establish reference behavior under fixed validation selection, not the superiority of a particular architecture.

Limitations. AcrossPlace inherits GSV-Cities' geographic biases and uses generated rather than human-authored descriptions. Source–target capture conditions differ, but median heading gaps are modest, so transfer gaps do not establish failure under extreme viewpoints. The stratified, single-reviewer audit cannot estimate dataset-wide error rates or inter-rater agreement. Place-ID recall does not certify textual specificity; three-epoch adaptations do not establish convergence; and cross-dataset transfer remains untested. Early P0 test feedback informed development, leaving independent confirmation on a hidden or external split important. Any data release must respect the source imagery's terms.

Conclusion. AcrossPlace establishes a reproducible standard for assessing whether language identifies places beyond the images used to describe them. Its principal protocols and diagnostic controls reveal when apparently comparable retrieval scores measure different observation, candidate, or language-evidence conditions.

Table 6. Matched 1,000-place gallery Recall@1 (%), averaged over five gallery draws and, when applicable, three training seeds. Other P0/P1 differences remain.

Reference	P0		P1	
	T2I	I2T	T2I	I2T
Pretrained BLIP	3.88	4.00	4.12	3.40
Pretrained SigLIP2	5.36	5.08	3.64	3.14
Supervised SigLIP2	13.58	11.69	10.61	9.37

Table 7. Blinded audit of 240 captions (120 per automatic stratum) by one reviewer. Unknown is distinct from contradiction or nonvisibility; counts are sample-specific.

Axis	Judgment	All	Valid	Uncertain
Source correctness	Supported	166	81	85
	Partial	8	2	6
	Contradicted	23	14	9
	Unknown	43	23	20
Target visibility	Visible	148	75	73
	Partial	9	2	7
	Not visible	37	17	20
	Unknown	46	26	20
Specificity	Distinctive	133	63	70
	Generic	42	23	19
	Unknown	65	34	31

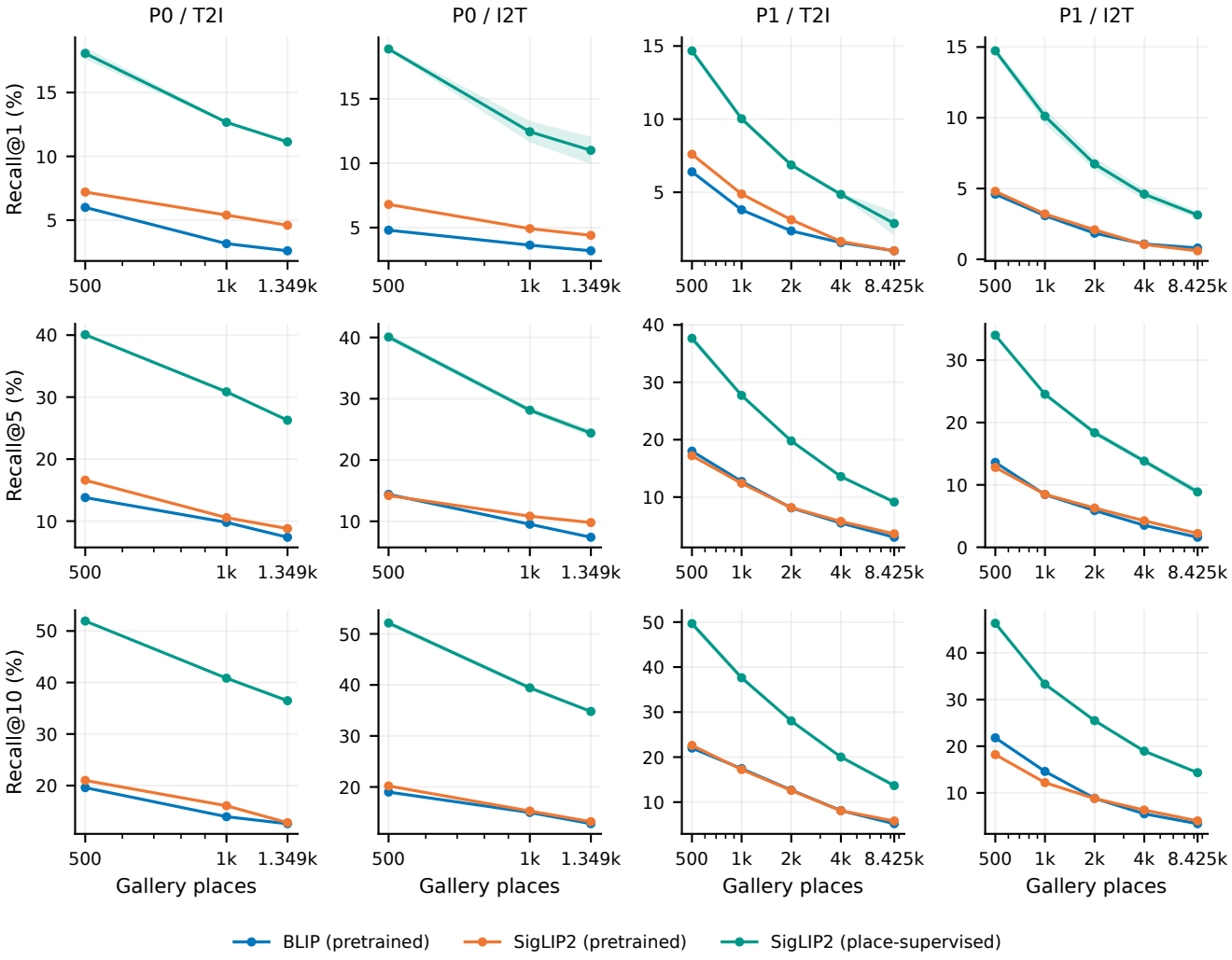


Figure 3. Fixed-query gallery-scale diagnostics across protocols, directions, and ranks. Each panel uses 500 query places and five nested galleries; supervised bands show seed SD. Full-pool repeats are identical; P0 and P1 end at 1,349 and 8,425 places.

References

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LLM Usage Statement

Qwen3-VL-4B-Instruct generated the dataset descriptions. Codex assisted implementation, analysis aggregation, source organization, and manuscript editing. A human reviewer performed the description audit. The authors are responsible for validating generated content, reported results, references, and final claims.